

NAME : RUSHIKESH GANESH TELANGADE
CLASS: TE(COMPUTER)
ROLL NO: 63
SECTION: C

Aim:- To perform data subsetting, sorting, and grouping operations to organize and analyse dataset attributes.

Source Code:-

```
import pandas as pd

df = pd.DataFrame({
    "Name": ["Arjun", "Kavya", "Sahil", "Meera"],
    "Dept": ["HR", "Sales", "Finance", "IT"],
    "Age": [24, 29, 26, 31],
    "Weight": [68, 55, 72, 60],
    "Salary": [35000, 42000, 38000, 50000]
})

print("Original Data:\n", df)

# Data Subsetting
print("\nSubset (Salary > 40000):\n", df[df["Salary"] > 40000])

# Sorting in Ascending Order
print("\nSorted by Salary (Ascending):\n",
      df.sort_values("Salary"))

# Sorting in Descending Order
print("\nSorted by Salary (Descending):\n",
      df.sort_values("Salary", ascending=False))

# Grouping
print("\nGrouped (Average Salary by Department):\n",
      df.groupby("Dept")["Salary"].mean())

# Statistical Operations
print("\nMean Salary =", df["Salary"].mean())
print("Median Salary =", df["Salary"].median())
print("Mode Salary =")
print(df["Salary"].mode())
```

Output:

Original Data:-

	Name	Dept	Age	Weight	Salary
0	Arjun	HR	24	68	35000
1	Kavya	Sales	29	55	42000
2	Sahil	Finance	26	72	38000
3	Meera	IT	31	60	50000

Subset (Salary > 40000):

	Name	Dept	Age	Weight	Salary
1	Kavya	Sales	29	55	42000
3	Meera	IT	31	60	50000

Sorted by Salary (Ascending):

	Name	Dept	Age	Weight	Salary
0	Arjun	HR	24	68	35000
2	Sahil	Finance	26	72	38000
1	Kavya	Sales	29	55	42000
3	Meera	IT	31	60	50000

Sorted by Salary (Descending):

	Name	Dept	Age	Weight	Salary
3	Meera	IT	31	60	50000
1	Kavya	Sales	29	55	42000
2	Sahil	Finance	26	72	38000
0	Arjun	HR	24	68	35000

Grouped (Average Salary by Department):

Dept	Average Salary
Finance	38000.0
HR	35000.0
IT	50000.0
Sales	42000.0

Name: Salary, dtype: float64

Mean Salary = 41250.0

Median Salary = 40000.0

Mode Salary =

0 35000

1 38000

2 42000

3 50000

Name: Salary, dtype: int64